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EST. 1837

Survival Analysis Assignment

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Introduction

For patients with end-stage renal disease, dialysis is a vital, life saving treatment. When kidney function is lost, it helps to preserve homeostasis. However,this treatment can have a detrimental effect on quality of life particularly in older patients, and is also linked to a high mortality rate. Conversely, conservative management, which emphasizes supportive medical care to manage symptoms, may be taken into consideration. This report, an assignment for the MSc in Biostatistics and Health Data Science Survival Analysis course, examines data from the esrd.csv dataset, a large cohort study of 8,796 elderly patients (aged 65 or older) with ESRD, contained in the esrd.csv dataset. Most patients in the study began Haemodialysis, while a smaller "control" group received conservative care, the majority of research participants started Haemodialysis. The time to death from any cause is the main study outcome. This analysis's main goal is to give a reliable estimate of how well Haemodialysis, as compared to conservative treatment, improves patient survival. The proportional decrease in mortality risk (aka the relative risk reduction) between the two groups is the precise definition of effectiveness. In order to address any confounding and produce a reliable final estimate, this analysis will first describe the dataset (Question 1) before proceeding through unadjusted and adjusted survival models.

Data Pre-processing: Handling Zero-Time Values

During the initial data check of the 'esrd.csv' dataset, it was observed that 24 patients had a recorded follow-up time (dthtime) equal to 0 (see Table 2). This corresponds to 10.9% (19/174) of the "Conservative" group and 0.06% (5/8622) of the "Haemodialysis" group.

Following the methodological recommendation provided, these patients **were not removed** from the analysis, as they represent the immediate baseline risk. However, to reflect a minimal follow-up time (e.g., < 1 day) and to ensure compatibility with all survival models (including the parametric models in Question 12 that require time > 0), all "dthtime = 0" values were replaced with a small, non-zero constant equal to half a day ($\frac{0.5}{365.25} \approx 0.00137$ years).

All subsequent analyses in this report are performed on this adjusted dataset.

Libraries used in code

In the code we are using the tidyverse package in addition to survival. The reason is that tidyverse provides a modern, highly readable, and efficient grammar for data manipulation and summarization. Functions like group_by() and summarise() allow for the creation of complex descriptive tables (like Table 1) in a single, auditable code pipeline, which is substantially clearer and less error-prone than using multiple nested base R calls like tapply(), table(), and prop.table().

Some more details about some **Tidyverse (dplyr)** key functions:

1. **mutate**: to create or modify variables, such as the time-dependent covariates
2. **filter**: to subset data
3. **select**: to choose specific columns for table
4. **group_by /summarise**: to aggregate descriptive statistics
5. **n=n()** : number of rows in each group

Additionally, the pipe operator (% > %) was used throughout the code to chain these operations into linear, readable workflows, passing the output of one function directly as the input to the next.

1 Question 1: Descriptive Statistics

Characteristic	Conservative	Haemodialysis	Total
N (%)	174 (2.0%)	8,622 (98.0%)	8,796 (100.0%)
Age (years), Mean (SD)	81.2 (7.2)	74.1 (5.8)	74.2 (5.9)
Sex, N (%)			
Female	165 (94.8%)	3,986 (46.2%)	4,151 (47.2%)
Male	9 (5.2%)	4,636 (53.8%)	4,645 (52.8%)
Comorbidities, N (%)			
COPD (Yes)	31 (17.8%)	741 (8.6%)	772 (8.8%)
Diabetes (Yes)	58 (33.3%)	3,149 (36.5%)	3,207 (36.5%)
Hypertension (Yes)	103 (59.2%)	5,408 (62.7%)	5,511 (62.7%)
Heart Disease (Yes)	122 (70.1%)	4,975 (57.7%)	5,097 (57.9%)
Liver Disease (Yes)	23 (13.2%)	479 (5.6%)	502 (5.7%)
Neoplasia (Yes)	27 (15.5%)	1,020 (11.8%)	1,047 (11.9%)
Vascular Disease (Yes)	89 (51.1%)	3,987 (46.2%)	4,076 (46.3%)
Follow-up Time (years), Mean (SD)	0.23 (0.73)	2.06 (2.10)	2.03 (2.10)
Event (Death), N (%)	71 (40.8%)	5,691 (66.0%)	5,762 (65.5%)

Table 1: Baseline characteristics of elderly patients with end-stage renal disease, by treatment modality.

	Conservative	Haemodialysis	Total
Patients with 0-day follow-up, n (%)	19 (10.9%)	5 (0.06%)	24 (0.27%)

Table 2: Patients who died or were censored at time zero.

The baseline characteristics of the 8,796 patients in the cohort, grouped by treatment, are shown in Table 1. As we can see there are significant and immediate disparities between the two groups. With only **174 (2%)** patients in the Conservative group and **8,622 (98%)** in the Haemodialysis group, the cohort is seemed to be highly unbalanced. These are also some notable demographic differences: the Conservative groups, on average, **7** years older than the Haemodialysis group (mean age **79.1** vs **72.1** years).

Moreover, the majority of members in the Haemodialysis group are men **(53.8%)**, while almost all members of the Conservative group are women **(94.8%)**. There is also a easily visible disparity in clinical characteristics that reflects this demographic mismatch. Major comorbidities known to impact survival, such as liver disease (**13.2% vs 5.6%**), COPD **(17.8% vs 8.5%)**, and heart disease **(70.1% vs 57.7%)**, are significantly more common in the Conservative group. Finally, there are differences in the actual observation period. The Conservative group's mean follow-up period was **0.23** years (SD 0.73), while the Haemodialysis group's was **2.06** years (SD 2.1). This is especially evident in the Conservative group's median follow-up time, which was only **0.04** years (about 15 days), suggesting a high frequency of early occurrences or fast loses to follow-up. As we can see in Table 2, **5,691 (66%)** of Haemodialysis group and **71 (40.8%)** of the Conservative group had passed away at the the end of the study.

In conclusion, there is no comparison between the two groups. The tough imbalances in age, sex and baseline comorbidities represent a classic case of confounding by indication. This means, that the treatment decision was obviously influenced by these initial patient characteristics. The data, strongly suggests, that a very specific small and sicker population (very old, mostly female, with high comorbidity) was selected for conservative management.

2 Question 2: Overall Death Risks & Crude Effectiveness

Table 3: Overall Death Risk (Cumulative Incidence) by Treatment Modality.

Treatment	Died (N)	Censored (N)	Total (N)	Death Risk
Conservative	71	103	174	0.408
Haemodialysis	5691	2931	8622	0.660
Total	5762	3034	8796	0.655

From the above Table 3 a crude estimate of effectiveness can be calculated as defined in the assignment. The Relative Risk (RR) is 1.62 (calculated as $\frac{0.660}{0.408}$), and the Relative Risk Reduction (RRR), or effectiveness, is -62% (calculated as $1 - RR$. So, it is way more than obvious that this analysis -incorrectly- suggests that Haemodialysis is significantly (Chi square test $\chi^2 = 47.94$ with p-value<0.05) associated with a 62% increase in the risk of death compared to the Conservative treatment group.

2.1 The reason this estimate is misleading

This is for three main reasons: This crude percentage-based analysis is statistically inaccurate and produces a highly misleading estimate of effectiveness.

1. **It Ignores Time:** This computation treats a patient who dies on the first day in the same way as a patient who dies five years later. The real survival time is not taken into consideration by this straightforward ratio.

2. **It Ignores Follow-up Duration:** The follow-up times were not equal, as shown in Table 1 compared to the Conservative group (mean **0.23** years), the Haemodialysis group was followed up for a significantly longer duration (mean **2.06** years). This indicates that a direct comparison of simple percentages is invalid because the Haemodialysis group had significantly more time "at-risk" for a death to be seen.

3. **It is Severely Confounded:** The two groups differ fundamentally in baseline characteristics (age, sex, comorbidities). This confounding by indication almost certainly explains the apparent excess risk in the Haemodialysis group rather than a true treatment effect.

These factors make crude cumulative incidence an insufficient indicator of treatment efficacy in this research. Next, we improve the analysis by using mortality rates per person-year to account for follow-up time.

3 Question 3: Mortality Rates per Person

As an improvement on the crude risk proportions, this section estimates the overall time-adjusted mortality rates, which account for the different follow-up times in each group by using a person-year denominator. The results are shown in Table 4.

Treatment	Total Deaths	Total Person-Years	Rate (per 100 PY)	95% CI
Conservative	71	39.8	178.4	[139.3, 225.0]
Haemodialysis	5691	17798.4	32.0	[31.1, 32.8]

Table 4: Overall Mortality Rates by Treatment Modality.

The analysis reveals a big difference in mortality rates. The conservative group had a mortality rate of **178.4 deaths per 100 person-years** (95% CI: $[139.3, 225.0]$, while the rate of the Haemodialysis group was **32 deaths per 100 person-years** (95% CI: $[31.1, 32.8]$. To confirm if this difference was significant we did a two sample poisson test that gave us highly statistical significant results with a **p-value** $<$ **0.001**.

This result is a complete reversal of the finding from Question 2. The crude risk proportion (which ignored time) suggested that Haemodialysis was 62% `worse`. By simply accounting for the fact that the Haemodialysis group had a much longer follow-up time, this rate-based analysis now suggests that Haemodialysis is 82.07% `better`. This suggests that the crude proportion method from Question 2 was extremely misleading.

Using these rates, the **estimated Rate Ratio (RR)** for Haemodialysis vs Conservative group is been found to be 0.179 ($\frac{32}{178.4}$).This results in a calculative effectiveness (Relative Rate Reduction) of **82.07%**, suggesting a very large protective effect from Haemodialysis treatment.

3.1 Is this an Optimal Approach

This is definitely not an optional approach. While it is indeed a significant improvement compared to Question 2 due to the fact that is correctly incorporates person-time, it still suffers from two major flaws that makes this **82.07%** reduction estimate completely unreliable:

1. **It assumes relatively constant Risks over time:** This method-dividing total events by total time-calculates an average rate over the entire study period. More specifically, it assumes that the risk of death for a patient is constant over time. This is for sure an unrealistic assumption, especially for elderly, sick patients whose risk is likely highest at the beginning of follow up.

2. **It is still confounded:** This is an **unadjusted** (crude) estimate. As we previously discussed in Question 1, we are comparing two totally different populations. The Conservative group is much older, mostly female, and has a much higher burden of baseline comorbidities. All these suggests as we previously said that this **82.07%** reduction is a reflection of this confounding, not a true treatment effect.

All the above suggests the need for a non-parametric method that does not assumes constant hazard such as the **Kaplan-Meier** that we are going to discuss in the next Question.

4 Question 4: Kaplan-Meier Survival

To provide a better, unadjusted comparison of survival, we move from simple rates to non-parametric survival analysis. A Kaplan-Meier survival plot was generated to visualize the survival probabilities over time for both treatment groups, as shown in Figure 1.

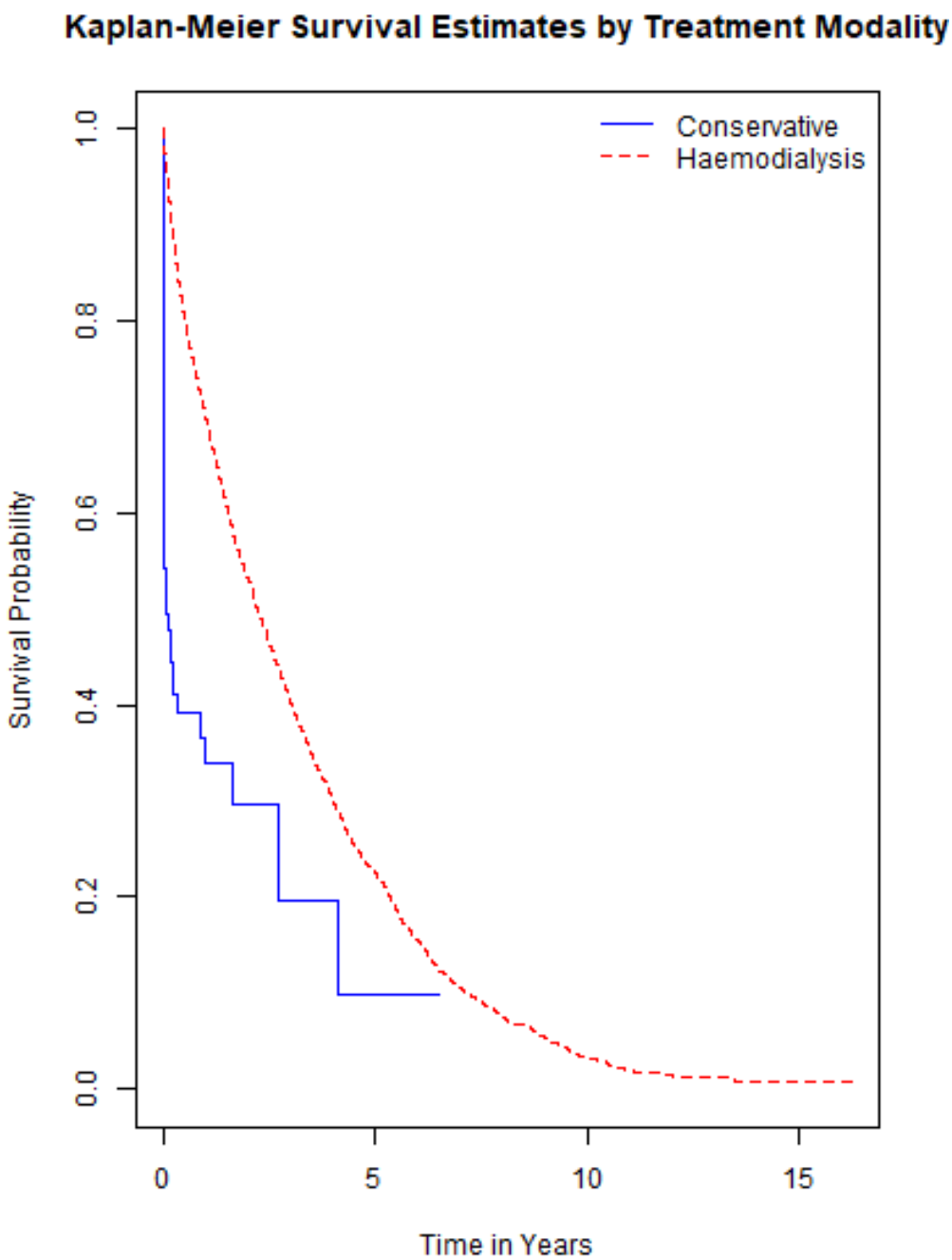

Figure 1: Kaplan-Meier survival estimates, stratified by treatment modality.

4.1 Interpret of the KM graph

The Figure 1 clearly shows fundamental difference in survival between the two groups. Over the course of the follow-up period, the survival (probability) curve is much higher for the **Haemodialysis** group (red line). This group's median survival is **2.24 years**.

On the other hand, the **Conservative** group (blue line) has a very low survival. With a median survival of just **0.093 years** (approximately 34 days), the curve declines extremely sharply.

- The **Haemodialysis group** (red line) has a significantly better survival probability throughout the entire follow-up period. The median survival for this group is **2.24 years**.
- The **Conservative group** (blue line) demonstrates extremely poor survival. The curve drops very steeply, with a median survival of only **0.093 years** (approximately 34 days).
- Compared to the (much) smoother curve for Haemodialysis,the Conservative curve's jagged, `steppy` appearance is a direct visual consequence of its extremely tiny sample size (n=174), where each individual death results in a significant, noticeable decline in the survival probability. The Haemodialysis curve ($n = 8,622$) consists of **5,691** tiny steps, which appear smooth to the eye.

4.2 Tests of significance (Log-Rank, Wilcoxon)

To examine if the difference observed in the graph (Figure 1) was significant, both the log-rank test and the Wilcoxon (Peto-Peto) test were applied.

- **Log-Rank test:** The test was highly significant ($p - value < 0.001$) suggesting reject of the null hypothesis that the two survival were identical.
- **Wilcoxon (Peto Peto) test:** The Wilcoxon test was also highly significant ($p{-}value < 0.001$)suggesting exactly the same as the Log-Rank test.

In conclusion, both tests are in strong agreement that a significant difference in survival exists between the two treatments groups (in favour of the Haemodialysis group). The Wilcoxon test puts more emphasis to early times, while the Log-Rank gives equal weight to all points. In this case, the Wilcoxon test statistic ($\chi^2 = 368$) is much larger than the log-rank statistic ($\chi^2 = 221$). This suggests that at the start of the follow-up the difference between the two groups is more pronounced.The above is in total agreement with our visual inspection of Figure 1, which displays the most dramatic separation between the two curves during the first year.

5 Question 5: Survival Probability Estimates

This section further quantifies the survival differences seen in the Kaplan-Meier plot by estimating specific survival probabilities at requested time points.

5.1 5(a) Estimated Survival Probabilities

The survival probabilities at 0.5, 1, 3, and 5 years were extracted from the Kaplan-Meier fit model from the previous section. The results are presented in Table 5.

Time (Years)	S(t) - Conservative	S(t) - Haemodialysis
0.5	0.393	0.806
1.0	0.339	0.702
3.0	0.198	0.406
5.0	0.099	0.225

Table 5: Estimated Survival Probabilities $S(t)$ at Specified Follow-up Times.

The table shows us in numbers the dramatic difference seen in the Kaplan-Meier plot. The survival probability for the Conservative group collapses extremely quickly, with only 39.3% of patients surviving the first 6 months and 33.9% surviving one year.

The Haemodialysis group seems to make it significantly better, with an **80.6%** survival probability in the first six months and **70.2%** at one year. Throughout the duration of time obviously the survival probability is starting to drop, but still, there is a significant **22.5%** of patients surviving in the Haemodialysis group at **5 years** mark, in contrast, with only **9.9%** of patients in the Conservative group at that specific time.

5.2 5(b) Conditional Survival Probabilities

This question asks for the conditional probability of surviving for an additional t years, given that the individual has already survived the first 6 months (0.5 years). This is mathematically written as $P(T > 0.5 + t \mid T > 0.5)$.

To estimate this, a new Kaplan-Meier model was fit only to the subset of patients who were alive and at-risk at $t = 0.5$ years. The time for these patients was re-baselined to $t - 0.5$. The estimated conditional survival probabilities are presented in Table 6.

Estimate	Add'l Time (t)	Cond. S(t) - Cons.	Cond. S(t) - Haem.
$S(t\|T > 0.5)$ at $t = 0.5yr$	0.5	0.862	0.871
$S(t\|T > 0.5)$ at $t = 1yr$	1.0	0.862	0.755
$S(t\|T > 0.5)$ at $t = 1.5yr$	1.5	0.754	0.661
$S(t\|T > 0.5)$ at $t = 2yr$	2.0	0.754	0.580

Table 6: Conditional Survival Probabilities for Patients Surviving > 6 Months (0.5 Years).

The Table 6 shows a fascinating turnaround and offers a prognosis for patients who make it through the initial high-risk phase. The survival probabilities for the first six months ($t = 0.5$) are almost the same (86.2% vs 87.1%) between Conservative and Haemodialysis group. From this point the trend seems to reverse from what we have previously observed. The Conservative group seems to have a better chance of surviving than

the Haemodialysis group for patients who have already lived for six months. To make this clear, we can take for example the probability of surviving one additional year ($t = 1.5$ total) is 75.4% for the Conservative survivors and just 66.1% for Haemodialysis survivors.

This is an tremendous example of a well known type of confounding called **survival treatment bias**. It suggests that the 21 patients in the Conservative group who made it through the first six months of high risk, must have been a very robust and healthy subset. Despite being much larger, the patient prognosis in the Haemodialysis group were nonetheless more diverse. This result strengthened the fact that these unadjusted comparisons are misleading and driven by extreme patient selection rather than actual treatment impact.

6 Question 6: Univariate Cox model for age

Following the initial non-parametric analyses, we continue our analysis by fitting a semi-parametric univariate Cox proportional hazard model to evaluate the impact of **age** on overall survival.

6.1 Assumption Checking: Linearity of Age

Before interpreting the model, we assessed the linearity assumption for the continuous variable `age` by examining the Martingale residuals. There were no significant non-linear patterns (such a quadratic U-shape) or extreme deviations in the smoothed plot of residuals vs age (Figure 2).The relation seemed sufficiently linear to support modeling age as a continuous linear predictor, despite some little variation at the extremities of the age distribution.

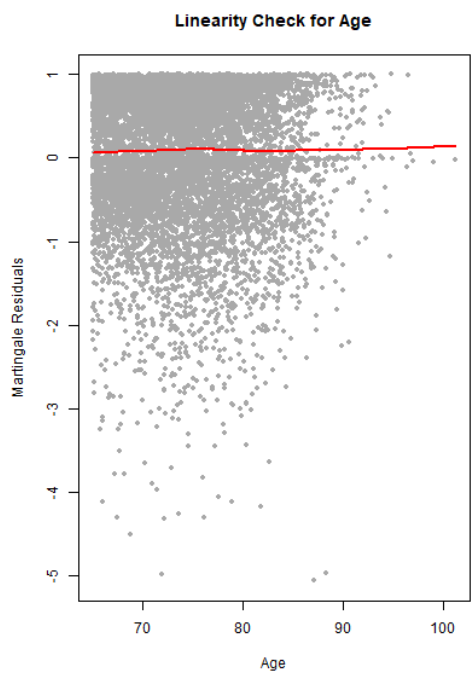

Figure 2: Martingale residuals vs. Age. The smoothed trend line (red) is approximately horizontal, supporting the assumption of linearity.

As shown in Figure 2, the smoothed trend line (red) remains relatively close to zero across the majority of the age range (up to 90 years). While there is some minor change around age 80 and a slight upward deviation

at the extreme upper ages (> 95), there is no distinct non-linear pattern (such as a U-shape). Therefore, the assumption of linearity is considered reasonably satisfied, and so, age is modeled as a continuous linear predictor.

6.2 Interpretation of the results

Variable	HR	CI	p.value
Age (per year)	1.033	[1.029,1.038]	< 0.001

Table 7: Univariate Cox Proportional Hazards Model for Age.

The output of the model fit is displayed in Table 7. We can see that age is a highly significant predictor of mortality in this cohort. The **HR** is 1.033 which means that for **every one-year increase** in a patient's age at baseline, their risk of death increases by 3.3%. The 95% CI is $[1.029, 1.038]$ confirms that age is indeed significant predictor, as the entire range is well above 1.

7 Question 7: Univariate Cox model for treatment

In this section we will evaluate the effect of the primary exposure (**treatment**) on survival using a univariate Cox PH model. In other words we will see an approximation of the unadjusted HR between the Conservative and Haemodialysis group following the semi-parametric cox model.

7.1 Interpretation of the results

Variable	Hazard.Ratio	CI	P.value
Haemodialysis (vs. Conservative)	0.2	$[0.158, 0.253]$	< 0.001

Table 8: Univariate Cox Proportional Hazards Model for Treatment.

The output of the model is displayed in Table 8. We can see from the table that **treatment** is a highly significant predictor on survival with a p-value$<$0.001. The **Hazard Ratio** for Haemodialysis is 0.2 (95% CI $[0.158, 0.253]$) suggesting that patients having the Haemodialysis treatment reduces the instantaneous risk of death by approximately 80% compared to conservative group. This is highly consistent with the crude rate ratio calculated in Question 3 (82.1%) and the large separation seen in the Kaplan-Meier plot (1) in Question 4.

However, it is important to note that this estimate is still **unadjusted**. Similar to the earlier analyses, this model ignores the fact that the Haemodialysis group is much younger and has a different comorbidity profile, attributing the entire survival difference to the treatment. As a result, even though this effectiveness estimate is statistically significant, it is probably skewed by serious confounding and should not be accepted until baseline covariates have been taken into account

8 Question 8: Checking Proportional Hazard Assumption

The validity of the Cox proportional hazards model relies on the assumption that the hazard ratio between the two groups remains constant over time. We used both formal and graphical methods to investigate whether this assumption is realistic for the treatment modality.

8.1 Formal Test: Grambsch & Therneau

The Grambsch and Therneau test, based on the scaled Schoenfeld residuals, was applied to the univariate Cox model. The output can be seen in Table 9.

Variable	Chi-squared	df	P-value
Treatment Modality	61.9	1	< 0.001
Global Test	61.9	1	< 0.001

Table 9: Test of Proportional Hazards Assumption (Grambsch & Therneau).

As we can see in the above Table 9, the test yielded a highly statistically significant result ($\chi^2 = 61.9$, p-value < 0.001). This provides strong statistical evidence to **reject the null hypothesis**, indicating a clear violation of the proportional hazards assumption.

8.2 Graphical Approach

Three graphical methods were used to visualize this violation.

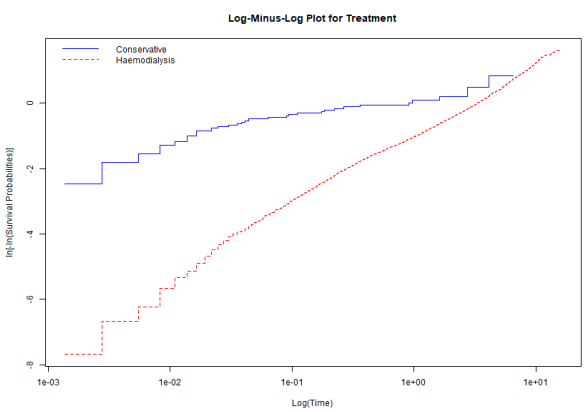

Figure 3: Log-Minus-Log (LML) survival plot.

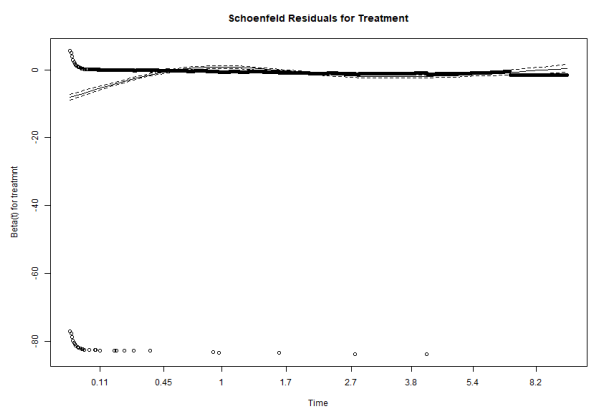

Figure 4: Scaled Schoenfeld residuals plot.

1. **Log-Minus-Log Plot (Figure 3):** Under proportional hazards, the curves for the two treatment groups should remain parallel across the log-time axis. It's way more than obvious though, that the curves are **not parallel**. They diverge greatly at first and converge over time, indicating that the relative risk is not constant but rather decreases with follow-up.

2. **Schoenfeld Residuals Plot (Figure 4):** The two treatment groups are represented by two separate bands of residuals in the plot. The trend of the coefficient over time is shown by the solid smoothed

line. The line shows a small upward movement, especially in the early era, even though it looks to be fairly horizontal to the eye. Even this slight trend is enough to cause the highly significant result in the formal test ($\chi^2 = 61.9$), given the enormous sample size (N=8,796). The positive slope implies the protective effect of treatment is weakening over time.

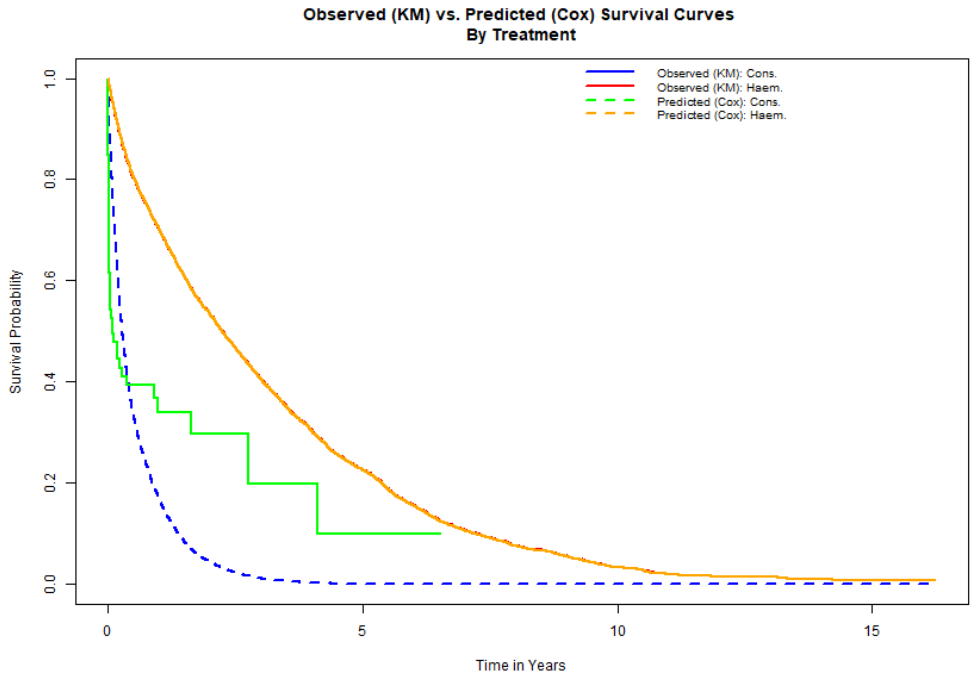

Figure 5: Observed (Kaplan-Meier) vs. Predicted (Cox) survival curves.

3. **Observed vs. Predicted Survival Curves (Figure 5):** This plot compares the raw non-parametric estimates (solid lines) with the semi-parametric estimates predicted by the Cox model (dashed lines).

   Visual inspection reveals a critical detail driven by the sample size imbalance. The Cox model fits the large Haemodialysis group (red) almost perfectly, as these 8,622 patients dominate the estimation. However, the model **fails to fit the Conservative group (blue)**. The observed survival (solid blue) drops much faster in the first year than the model predicts (dashed blue). The model, constrained by the assumption of a single constant hazard ratio, essentially averages the risk, failing to capture the specific high-risk profile of the smaller Conservative group.

8.3 Conclusion on PH Assumption

Based on the highly significant **Grambsch & Therneau** test and the consistent evidence from all three graphical methods we discussed above, the **proportional hazards assumption is violated**. The formal test and the cloglog plot verify that the hazard ratio is not constant, even though the Schoenfeld plot indicates the deviation is not visually dramatic. In particular, the protective impact of Haemodialysis seems to be greatest during the initial follow-up period and gradually fades.

9 Question 9: Testing PH Assumption via Time-Dependent Variable

The **Proportional Hazards (PH)** assumption was formally **rejected in Question 8**. To confirm this finding and model the nature of the time dependency, we fit a Cox model including a time-dependent variable (TDC) for treatment. The TDC multiplies the treatment effect by a function of time, $\log(t)$, allowing us to test if the Hazard Ratio changes systematically over the follow-up period.

The model fitted is:

$$\ln(h(t)) = ln(h_0(t)) + \beta_{trt} \cdot \text{treatmnt} + \beta_{TDC} \cdot \text{treatmnt} \cdot \ln(t)$$

This is implemented in R using the `tt()` function.

The key output is the coefficient associated with the time-dependent term, treatmnt:log(Time). The results are presented in Table 10.

Variable	Coefficient	95% CI	p-value
Haemodialysis	0.686	$[0.018, 1.354]$	0.044266
Haemodialysis: $log(Time)$	0.933	$[0.764, 1.103]$	< 0.001

Table 10: Cox Model with Time-Dependent Covariate to Test PH Assumption.

9.1 Interpretation of Results

The findings offer solid proof of the PH assumption:

- **Significance:** The coefficient for the time-dependent term (**Haemodialysis: log(Time)**) is highly statistically significant (**p-value** < 0.001).
- **Direction:** The coefficient is positive ($\beta_{TDC} = 0.933$), with a 95% Confidence Interval of [0.764, 1.103] that does not cross zero.

Because the time-dependent term is significant, we **strongly reject the null hypothesis** of proportional hazards. This confirms the conclusion from **Question 8** that the hazard ratio is not constant.

The positive coefficient ($\beta_{TDC} = 0.933$) indicates that the log-hazard ratio increases substantially as time progresses.

- At very early times (where $\log(t)$ is large and negative), the total effect ($\beta_{trt} + \beta_{TDC} \cdot \log(t)$) is strongly negative, indicating a massive protective effect of Haemodialysis.
- As time increases (and $\log(t)$ approaches 0 or becomes positive), this protective effect rapidly fades.

This confirms that the survival advantage of Haemodialysis is heavily concentrated in the early follow-up period.

10 Question 10: Addressing PH Violation via Piecewise Cox Model

The primary objective of this study is to estimate the effectiveness of Haemodialysis in improving patient survival. However, the significant violation of the proportional hazards assumption (Question 8) implies that the "effectiveness" is not constant, but changes over time. To provide a valid estimate that respects this dynamic, we implemented a piecewise Cox model as described in Collett's book, section 11.2, pages 336-339 (4th ed.).

Based on the visual evidence from the diagnostic plots (Figure 3) and the crossing survival probabilities presented in Question 5(b), we modeled the treatment effect separately for the **early phase (0–1 year)** and the **late phase (1+ years)**. The results are presented in Table 11.

Time Interval	Hazard Ratio	95% CI	Effectiveness (RRR)
0-1 Year	0.136	[0.107, 0.174]	86.4%
1+ Years	1.628	[0.525, 5.051]	-62.8%

Table 11: Piecewise Cox Model: Hazard Ratios and 95% Confidence Intervals by Time.

10.1 Interpretation in Relation to Study Objective

The piecewise model reveals that the "effectiveness" of haemodialysis is highly time-dependent:

1. **Early Effectiveness (0–1 Year):** In the first year, Haemodialysis is extremely effective. With a Hazard Ratio of **0.136** (95% CI: 0.107–0.174), the risk of death is reduced by **86.4%** when compared to conservative group. This captures the period where the conservative group experiences high acute mortality.
2. **Late Effectiveness (1+ Years):** For patients who survive the first year, the effect reverses. The 95% CI: (0.525–5.051) is quite wide and includes 1, showing low precision because of the limited number of conservative survivors, despite the point estimate suggesting a potential reversal of the effect. This is probably due to residual confounding and severe selection effects.

Conclusion

Reporting a single effectiveness estimate (like 80%) is invalid for this population. Long-term survivors do not benefit from the significant survival advantage that Haemodialysis provides during the crucial first year. These estimates are still unadjusted, though. The reversal in the second period likely reflects survivor bias and baseline confounding (e.g., the conservative survivors being a distinct "healthy" subset), reinforcing the need for the multivariable adjustment in Question 11.

11 Question 11: Multivariable Cox Model & Diagnostics

To provide a valid estimate, we fitted a multivariable model adjusting for all baseline confounders (age, sex, comorbidities).

11.1 Assessment of Interactions

We systematically examined the possibility of effect modification by fitting a Cox model that included two-way interactions between treatment modality and each of the baseline covariates. The results are shown in Table 11.1.

Interaction	Hazard Ratio	95% CI	p-value
Treatment x age	1.013	$[0.975, 1.052]$	0.502
Treatment x sex	6.477	$[1.921, 21.843]$	0.003
Treatment x copd	0.700	$[0.344, 1.424]$	0.324
Treatment x dm	1.307	$[0.677, 2.525]$	0.425
Treatment x hypert	7.102	$[3.828, 13.177]$	< 0.001
Treatment x heart	0.239	$[0.125, 0.459]$	< 0.001
Treatment x liver	0.964	$[0.431, 2.156]$	0.928
Treatment x neoplasia	0.393	$[0.209, 0.739]$	0.004
Treatment x vascular	0.563	$[0.308, 1.032]$	0.063

Table 12: Test for Interactions between Treatment and Baseline Covariates.

- **Results:** As seen in Table 11.1, several interactions are statistically significant, more specifically sex ($p = 0.003$,the CI $[1.9, 21.8]$), hypertension ($p < 0.001$, CI $[3.83, 13.2]$), and heart disease ($p < 0.001$. CI $[0.125, 0.46]$).
- **Methodological Decision:** We decided not to include these interactions in the final efficacy model, even though they were statistically significant. This decision was made cause of the severe sparsity of data in the Conservative control group. As noted in the descriptive analysis, the Conservative group contained only 9 male patients. Estimates such as the Hazard Ratio of 6.47 for the "Treatment x Sex" interaction are likely unstable outcomes of this small sample size rather than true clinical effects.
- **Conclusion:** To provide a robust and interpretable estimate of the overall effectiveness of Haemodialysis, and to avoid overfitting to small, unstable subgroups, we proceeded with the main effects model adjusted for baseline confounders.

11.2 Goodness of Fit Assessment (Cox-Snell Residuals)

The overall fit of the multivariable Cox model was assessed using Cox-Snell residuals. If the model fits the data well, the true cumulative hazard of the residuals should equal the residuals themselves, implying that the estimated cumulative hazard plot should follow a straight line with unit slope.

As shown in Figure 6, the estimated cumulative hazard of the residuals (jagged line) tracks the reference line reasonably well for the majority of the follow-up period. While there is some deviation in the right-hand tail (representing long survival times), this is expected due to the reduced number of patients at risk. Overall, the plot indicates that the multivariable Cox model provides an adequate fit to the data structure.

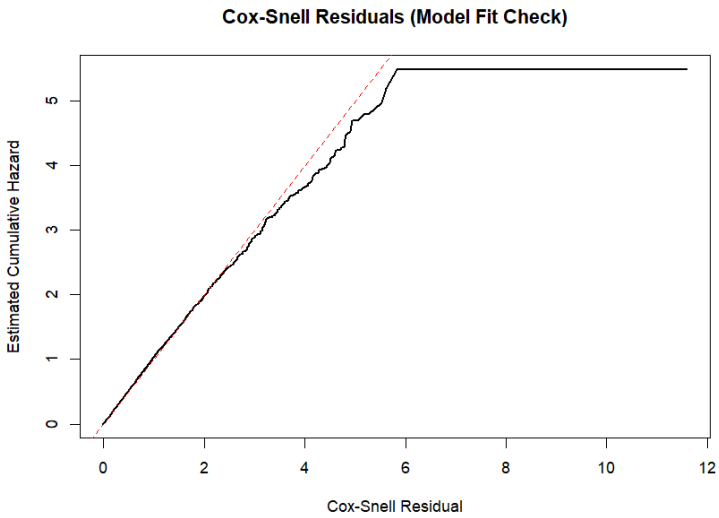

Figure 6: Cox-Snell residuals plot for the multivariable Cox model. The red dashed line represents perfect fit ($y = x$).

11.3 Hazards Re-assessment:

Before accepting the final multivariable model, we re-examined the proportional hazards assumption to see if adjusting for confounders had corrected the violation observed in the univariate analysis.

The **Grambsch & Therneau** test as displayed in Table 13 confirms that the proportional hazards assumption is violated for the global model **(p< 0.001)**. Specifically, **Treatment**, **Diabetes**, and **Hypertension** show highly significant deviations. This confirms that adjusting for baseline covariates did not resolve the time-dependent nature of the treatment effect, validating the need for the final Piecewise Cox model.

Covariate	Chi.squared	P.value
treatment	58.56	< 0.001
age	1.93	0.164594
sex	4.89	0.027067
copd	1.35	0.245542
dm	11.17	< 0.001
hypert	51.33	< 0.001
heart	0.29	0.587813
liver	2.46	0.116662
neoplasia	1.32	0.250543
vascular	3.68	0.055119
Global Test	137.91	< 0.001

Table 13: Test of Proportional Hazards Assumption (Multivariable Model).

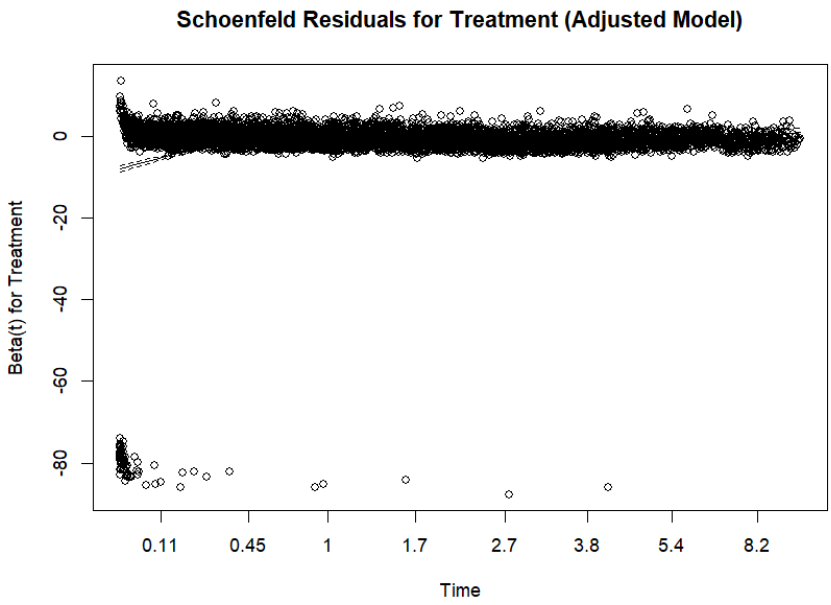

Figure 7: Scaled Schoenfeld residuals for Treatment in the multivariable model. The solid line represents the smoothed trend of the coefficient over time.

Visual Interpretation: The figure shows two different bands of residuals that correspond to the two treatment groups, as shown in Figure 7. The formal test finds a substantial trend, which is probably caused by the lower band (the Conservative group) or slight drifts in the bigger group, even though the upper band (the huge Haemodialysis group) looks rather flat. The final Piecewise Cox model is necessary since the substantial p-value demonstrates that the treatment coefficient is not constant over time.

11.4 Final Model

To address the PH violation while adjusting for confounding, we extended the Piecewise Cox model from Question 10 to include all of the baseline covariates. The results are shown in Table 14.

Time Interval	Adjusted HR	95% CI	Adj. Effectiveness (RRR)
0-1 Year	0.153	[0.119, 0.195]	84.7%
1+ Years	1.798	[0.578, 5.592]	-79.8%

Table 14: Multivariable Adjusted Effectiveness of Haemodialysis (Piecewise Cox Model).

Conclusion

After ajusting for baseline confounders, the time-dependent nature of the treatment effect remains still:

- **Early Phase (0–1 Year):** Haemodialysis provides a substantial and statistically significant survival benefit. In comparison to conservative management, the Adjusted Hazard Ratio is **0.153** (95% CI: $[0.119, 0.195]$), which indicates a **84.7% reduction** in the risk of death. This demonstrates how successful the medication is at preventing early death.

- **Late Phase (1+ Years):** The Adjusted Hazard Ratio increases to **1.798** after a year, indicating a possible reverse where Haemodialysis may be linked to increased risk. Nevertheless, the 95% Confidence Interval contains 1.0 and is incredibly broad $[0.578, 5.592]$. This suggests that the late-period effect is **not statistically significant**.

Overall Assessment: The analysis shows that the effectiveness of Haemodialysis is heavily concentrated in the first year of follow-up. Due to the low number of survivors in the Conservative group following the significant early mortality, the apparent "harm" in the later time is estimated with very low precision. As a result, we can say with confidence that Haemodialysis provides a significant early survival advantage, but this evidence raises questions about in long term efficacy in comparison to conservative treatment.

12 Question 12: Parametric Weibull Modeling

As an alternative to the semi-parametric Cox model, we explored the use of a fully parametric Weibull model.

12.1 (a) Assessment of Fit and Parameter Estimation

To check if the Weibull distribution is a good fit to the data, we employed two graphical approaches: comparing the predicted survival curves to the observed data (Figure 8) and examining the linearity of the log-cumulative hazard (Figure 9)-it's mathematically the same plot as in Figure 3 of Question 8 but we will show it again for convenience-.

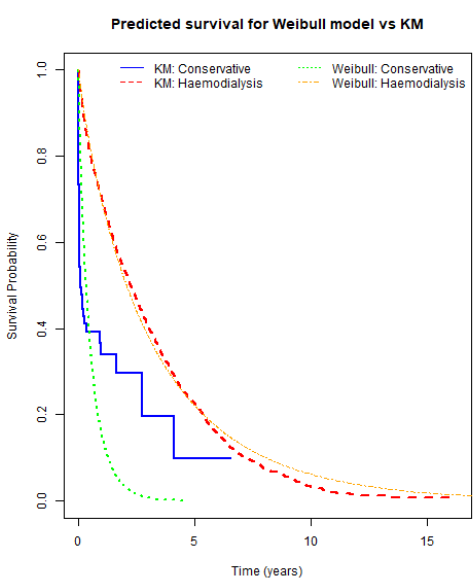

Figure 8: Observed (KM) vs. Weibull Predicted survival curves.

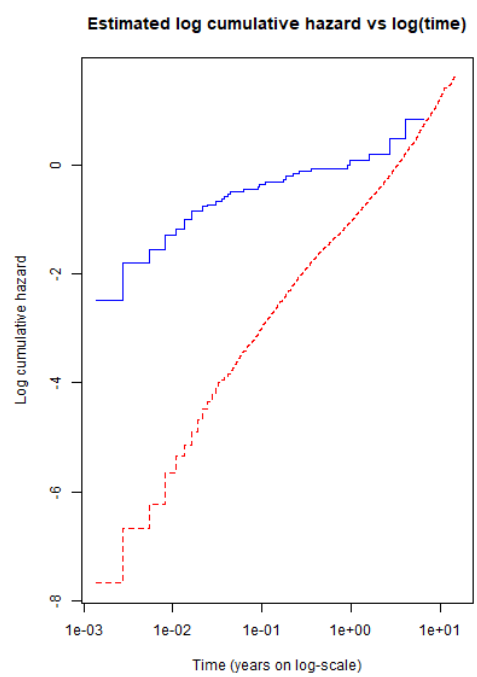

Figure 9: Log-Cumulative Hazard Plot. Non-parallel lines indicate unequal shape parameters.

Goodness of Fit:

Figure 8 illustrates how the Weibull model's suitability differs by group:

- **Haemodialysis Group:** For much of the follow-up period time, the Weibull prediction (orange dashed line) tracks the observed Kaplan-Meier curve (red solid line) quite well, indicating that the distribution is a suitable approximation for this group.
- **Conservative Group:** In the later time, the fit clearly deviates.While the Weibull model (green dashed line) correctly follows the rapid initial decline down to a survival probability of approximately 0.4, it fails to capture the behaviour of the "tail". A subset of long-term survivors is suggested by the observed data (blue solid line), which levels off at about 10% survival.The Weibull model overestimates late mortality for this population since it is limited by its parametric form and keeps trending lower toward zero.

Figure 9 confirms this lack of fit diagnostically. These plots ought to be straight lines for a Weibull distribution. There is a noticeable curvature in the Conservative group (blue). Additionally, the lines are not parallel, which defies the conventional Weibull model assumptions by showing that the "Shape" parameter is not shared by the groups.

Conclusion on Fit:

While the Weibull distribution captures the general trend of high early mortality, it does not perfectly describe the survival experience of the Conservative group, especially the long-term stabilization. However, we continued with the graphical and Maximum Likelihood Estimation (MLE) comparison in order to assess the parameter estimates.

Parameter Estimation: We visually assessed the parameters from Figure 9 and contrasted these qualitative impressions with the Maximum Likelihood Estimates (MLE).The results are shown in Table 15

Parameter	MLE Estimate	95% CI
Shape (p)	0.878	[0.860, 0.897]
Scale (alpha) - Cons.	0.494	[0.378, 0.644]
Scale (alpha) - Haem.	3.121	[3.030, 3.215]

Table 15: MLE Weibull Parameters

Contrast: The MLE "Shape" parameter of **0.88** represents the best-fitting *average* slope under a model that constrains both treatment groups to share a common Weibull form parameter. On the other hand, heterogeneity is evident in the graphical diagnosis in Figure 9. The Conservative curve is notably flatter and shows curvature at later dates, while the Haemodialysis curve has a steeper, almost linear pattern. This suggests that the underlying hazard pattern is different for the two groups. As a result, the Weibull model is forced to choose between two distinct behaviors when a single Shape parameter of 0.88 is imposed, which results in a poor depiction of the survival tail of the Conservative group.

12.2 (b) Multivariable Weibull Model Predictions

We fitted a multivariable Weibull model adjusting for the same baseline covariates as the Cox model in Question 11. Figure 10 displays the predicted survival curves for a representative patient (Male, mean age, with hypertension and heart disease) in each treatment group.

Comparison with Cox Model (Q11): Qualitatively, the Weibull model produces smooth, monotonic curves that show a consistent survival advantage for Haemodialysis (red) over Conservative management (blue) throughout the entire follow-up period.

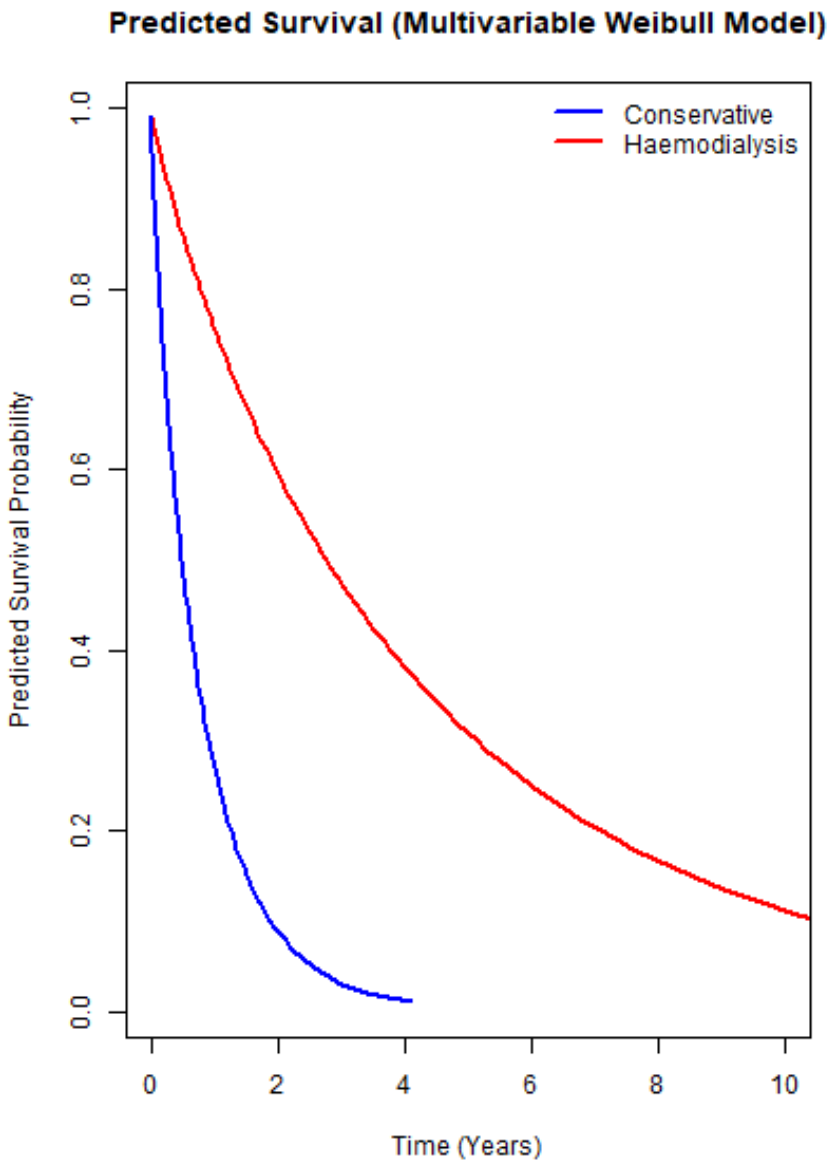

Figure 10: Predicted survival curves from the multivariable Weibull model.

This partially agrees with the "Early Phase" result from the Piecewise Cox model but **disagrees** with the "Late Phase" finding. The standard Weibull model forces a specific parametric shape onto the data (Accelerated Failure Time). It cannot easily model the complex time-dependent dynamic we found in Question 11, where the massive early protective effect diminishes or even reverses after the first year. The Weibull model essentially "averages" these effects into a single optimistic curve for the Haemodialysis group.

Conclusion: Would we use the Weibull model? No. We would not recommend using the standard Weibull model for this dataset.

1. **Fit Issues:** As shown in 12(a), the model fails to capture the specific hazard shape (steep initial drop) of the Conservative group.

2. **Assumption Violation:** The diagnostic plots confirmed that the "Shape" parameters differ between groups, violating the proportional hazards assumption inherent in the standard Weibull model.

3. **Clinical Insight:** Most importantly, the rigid parametric form obscures the critical finding that the treatment effectiveness diminishes over time. The Piecewise Cox model (Question 11) provides a more flexible and clinically accurate description of the changing risk profile.